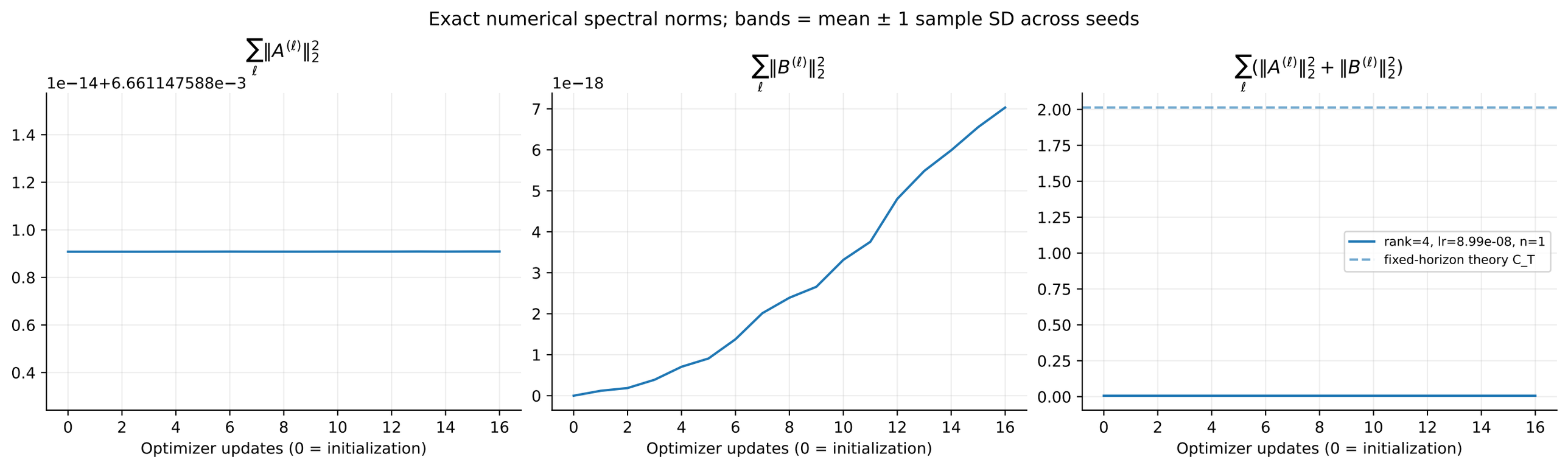
Exact numerical spectral norms; bands = mean ± 1 sample SD across seeds
$\sum_\ell \|A^{(\ell)}\|_2^2$
1e−14+6.661147588e−3
1.4
1.2
1.0
0.8
0.6
0.4
0
2
4
6
8
10
12
14
16
Optimizer updates (0 = initialization)
$\sum_\ell \|B^{(\ell)}\|_2^2$
1e−18
7
6
5
4
3
2
1
0
0
2
4
6
8
10
12
14
16
Optimizer updates (0 = initialization)
$\sum_\ell (\|A^{(\ell)}\|_2^2 + \|B^{(\ell)}\|_2^2)$
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
rank=4, lr=8.99e-08, n=1
fixed-horizon theory C_T
0
2
4
6
8
10
12
14
16
Optimizer updates (0 = initialization)